HOME / I MIEI CORSI / DATA MINING M - MACHINE LEARNING (A.A. 19/20) / SEZIONI / M1 - MACHINE LEARNING EXAM / MACHINE LEARNING - MULTIPLE CHOICES TEST

Iniziato	martedì, 7 gennaio 2020, 15:06
Stato	Completato
Terminato	martedì, 7 gennaio 2020, 15:36
Tempo impiegato	30 min. 1 secondo
Punteggio	15,00/15,00
Valutazione	**30,00** su un massimo di 30,00 (**100**%)

Domanda **1**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

How does *pruning* work when generating frequent itemsets?

Scegli un'alternativa:

a. If an itemset is frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated

b. If an itemset is frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated

c. If an itemset is not frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated

d. If an itemset is not frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated

Domanda **2**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which is the main reason for the *standardization* of numeric attributes?

Scegli un'alternativa:

a. Map all the numeric attributes to a new range such that the mean is zero and the variance is one.

b. Remove non-standard values

c. Map all the nominal attributes to the same range, in order to prevent the values with higher frequency from having prevailing influence

d. Change the distribution of the numeric attributes, in order to obtain gaussian distributions

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following statements regarding the discovery of association rules is true? (One or more)

Scegli una o più alternative:

NO a. The support of a rule can be computed given the confidence of the rule

b. The confidence of a rule can be computed starting from the supports of itemsets

c. The confidence of an itemset is anti-monotonic with respect to the composition of the itemset

d. The support of an itemset is anti-monotonic with respect to the composition of the itemset

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Given the two binary vectors below, which is their similarity according to the Jaccard Coefficient?

a b c d e f g h i j

1 0 0 0 1 0 1 1 0 1

1 0 1 1 1 0 1 0 1 0

Scegli un'alternativa:

a. 0.5

b. 0.2

c. 0.1

d. 0.375

non-match

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is the *cross validation*

Scegli un'alternativa:

a. A technique to obtain a good estimation of the performance of a classifier with the training set

b. A technique to obtain a good estimation of the performance of a classifier when it will be used with data different from the training set

c. A technique to improve the speed of a classifier

No d. A technique to improve the quality of a classifier

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is different from the others?

Scegli un'alternativa:

a. Decision Tree

b. SVM

c. Dbscan

d. Neural Network

Domanda **7**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is the main purpose of *smoothing* in Bayesian classification?

Scegli un'alternativa:

a. Dealing with missing values

b. Classifying an object containing attribute values which are missing from some classes in the training set

c. Reduce the variability of the data

d. Classifying an object containing attribute values which are missing from some classes in the test set

Domanda **8**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

The *information gain* is used to

Scegli un'alternativa:

a. select the attribute which maximises, for a given test set, the ability to predict the class value

b. select the attribute which maximises, for a given training set, the ability to predict all the other attribute values

c. select the attribute which maximises, for a given training set, the ability to predict the class value

d. select the class with maximum probability

Domanda **9**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which of the following *is not* an objective of feature selection

Scegli un'alternativa:

a. Avoid the *curse of dimensionality*

b. Reduce time and memory complexity of the mining algorithms

c. Select the features with higher range, which have more influence on the computations

d. Reduce the effect of noise

Domanda **10**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which of the statements below is true? (One or more)

Scegli una o più alternative:

a. K-means is very sensitive to the initial assignment of the centers
the number of attributes is very large k-means is prone to the *curse o*

b. Sometimes k-means stops to a configuration which does not give the minimum distortion for the chosen value of the number of clusters.

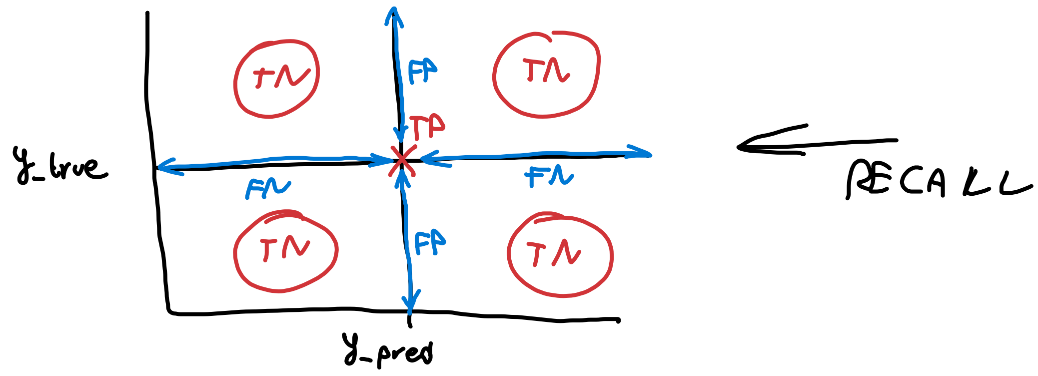
c. K-means is quite efficient even for large datasets
distortion for an assigned number of clusters

d. K-means always stops to a configuration which gives the minimum distortion for the chosen value of the number of clusters.

Domanda **11**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Given the definitions below:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives



which of the formulas below computes the *precision* of a binary classifier?

PRECISION

Scegli un'alternativa:

a. TP / (TP + FN)

b. (TP + TN) / (TP + FP + TN + FN)

✓ ~~c.~~ TP / (TP + FP)
positives divided b

d. TN / (TN + FP)

Domanda **12**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which of the following clustering methods is ***not*** based on distances between objects?

Scegli un'alternativa:

a. DBSCAN

b. Hierarchical Agglomerative

✓ ~~c.~~ Expectation Maximization

d. K-Means

Domanda **13**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

In a dataset with D attributes, how many subsets of attributes should be considered for feature selection according to an exhaustive search?

Scegli un'alternativa:

a. O(D)

✓ ~~b.~~ $O(2^D)$

c. O(D!)

d. $O(D^2)$

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

After fitting DBSCAN with the default parameter values the results are: 0 clusters, 100% of noise points. Which will be your next trial?

Scegli una o più alternative:

√ a. Reduce the minimum number of objects in the neighborhood

b. Reduce the minimum number of objects in the neighborhood and the radius of the neighborhood

c. Decrease the radius of the neighborhood

√ d. Increase the radius of the neighborhood

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a decision tree, an attribute which is used only in nodes near the leaves...

Scegli un'alternativa:

a. ...has a high correlation with respect to the target

b. ...is irrelevant with respect to the target

c. ...guarantees high increment of purity

√ d. ...gives little insight with respect to the target

◄ Lab Activity 17-12-2019 - Simulation of lab exa Vai a... Introduction to Big Data - Slides ►